

HOLA: Towards Holistic Human–AI Collaborative Framework for Interactive Aortic CTA Segmentation and Quality Assessment

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Abstract. Accurate segmentation of the aorta from computed tomography angiography (CTA) is essential for diagnosis, treatment planning, and quantitative assessment of aortic disease. Manual segmentation is time-consuming and subject to inter-observer variability, while fully automated models can produce errors that are difficult to correct in complex anatomy. We propose HOLA, a three-dimensional (3D) human-AI collaborative framework combining automatic segmentation, interactive volumetric correction, and quality-aware feedback. HOLA integrates a 3D DynUNet with the DeepEdit interaction strategy, using positive and negative click maps to iteratively refine segmentations across the full CTA volume, trained via a progressive multi-dataset strategy across six CTA datasets to improve generalisation across heterogeneous anatomy and pathology. On 72 held-out test cases, mean Dice improved from 0.804 to 0.868 after ten clicks (Wilcoxon signed-rank, $p < 0.001$). A quality prediction network estimated segmentation quality without ground truth ($r = 0.953$, MAE = 0.039), enabling a quality-guided stopping criterion achieving 95.3% precision while reducing interaction from 10.00 to 1.03 clicks per case. These results demonstrate the potential of 3D interactive refinement with reference-free quality assessment for efficient human-AI collaborative aortic segmentation.

Keywords: Aortic Segmentation · Computed Tomography Angiography · Interactive Segmentation · Human–AI Collaboration · Deep Learning · Quality Assessment · Aortic Disease

1 Introduction

Recent advances in deep learning have significantly improved automated medical image segmentation. Early convolutional neural network (CNN) architectures, particularly

U-Net [13], demonstrated that encoder-decoder networks with skip connections could accurately capture anatomical structures across multiple imaging tasks. This was followed by nnU-Net [5], which automated network configuration and preprocessing to achieve state-of-the-art performance without extensive manual optimisation. More recently, transformer-based architectures have improved volumetric segmentation by modelling long-range spatial relationships within three-dimensional images. Methods such as UNETR [3] and CIS-UNet [4] further enhanced segmentation accuracy for complex vascular structures by combining convolutional representations with attention mechanisms. Foundation models including SAM [8], Medical SAM 2 [17], and MedSAM2 [9] have extended segmentation to prompt-guided interaction across different imaging modalities and anatomical structures. Despite these advances, they lack aorta-specific training and a mechanism to assess segmentation quality.

Interactive segmentation has emerged as a practical alternative by allowing users to iteratively refine segmentation results through simple corrective interactions. Rather than replacing the clinician, interactive approaches maintain human involvement throughout the segmentation process, making them particularly suitable for anatomically complex cases where automatic methods often fail. DeepGrow [14] first demonstrated that positive and negative click maps could guide segmentation refinement, while DeepEdit [1] unified automatic segmentation and interactive correction within a single learning framework. These methods enable segmentation errors to be corrected efficiently without repeating the entire annotation process, improving both flexibility and clinical usability.

Despite these advances, existing approaches remain difficult to integrate into routine clinical practice. Most automatic methods generate a single fixed segmentation without an effective mechanism for correction, while current interactive approaches provide limited support for volumetric refinement and rarely indicate whether the resulting segmentation is sufficiently accurate for clinical use.

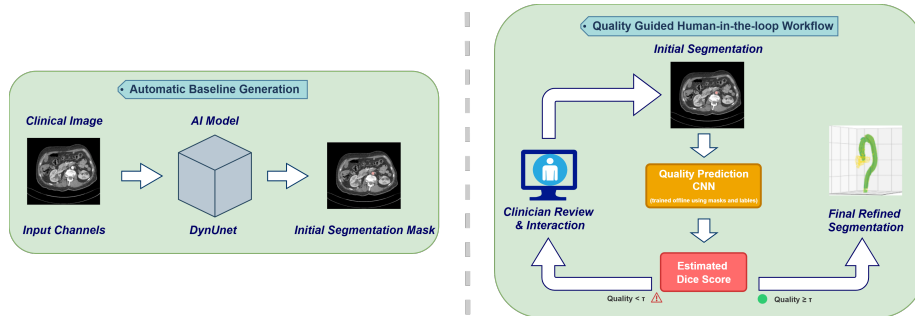


Fig. 1. Comparison between Conventional Automatic Segmentation and the Proposed Quality-guided Human-in-the-loop Segmentation Framework

To address these challenges, HOLA introduces an end-to-end interactive segmentation framework covering the complete loop—from initial AI prediction and human

correction to updated prediction and quality estimation. Moving beyond basic component integration, positive and negative click maps are incorporated directly into a 3D DynUNet backbone, allowing fully volumetric refinement across the entire CTA volume rather than individual slices. To improve robustness, the framework uses a progressive multi-dataset strategy incorporating different aortic pathologies sequentially. Additionally, a quality prediction module predicts segmentation quality directly from the generated mask without requiring ground-truth annotations during inference. This predicted quality score feeds into an interactive Streamlit interface [15], providing visual feedback and aiming to improve segmentation accuracy, model robustness, and practical deployment.

2 Methodology

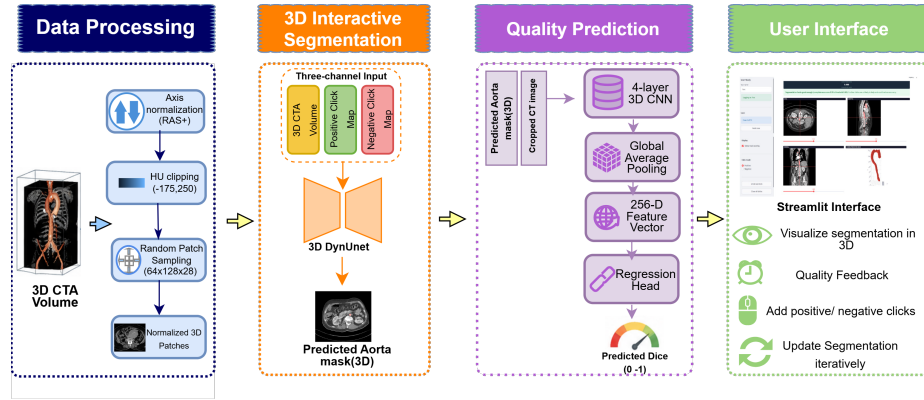


Fig. 2. Workflow of HOLA Interactive Segmentation Framework

HOLA consists of four integrated stages: CTA preprocessing, 3D interactive segmentation, automated quality prediction, and user-guided refinement. Unlike conventional automatic segmentation approaches that generate a single static prediction, HOLA enables interactive refinement with automated quality assessment to support clinical decision-making.

The workflow begins with a 3D CTA scan, which is first preprocessed to standardise image orientation and intensity values. During interactive segmentation, the processed CTA volume is combined with positive and negative click maps generated from user interactions. These three channels are provided as input to a DeepEdit-enabled 3D DynUNet [1], which produces an initial segmentation and iteratively refines the prediction using user corrections. To assess segmentation reliability, a lightweight 3D convolutional neural network predicts the expected quality score directly from the predicted mask and its local CT context without requiring ground-truth annotations during inference. The

estimated quality score provides an objective indication of segmentation reliability and is compared with a predefined stopping threshold to determine whether additional user corrections are likely to improve the result. Finally, the predictions and quality scores are displayed through an interactive Streamlit interface [15], allowing users to review the segmentation, receive visual feedback, and refine the result when necessary.

2.1 Dataset Preparation and Preprocessing

HOLA was developed using six CTA datasets, including normal anatomy and several aortic pathologies. The datasets comprise the original baseline cohort together with AortaSeg-60 [2], ImageTBAD [16], AVT (Seg.A) [11], CIS-UNet [4], and the Aortic Dissection Dataset and Segmentations [10]. The segmentation model was trained sequentially by progressively incorporating additional datasets from the smallest to the largest patient cohort (Base $\rightarrow$ Dissection [10] $\rightarrow$ SEGA [11] $\rightarrow$ CIS-UNet [4] $\rightarrow$ AortaSeg-60 [2] $\rightarrow$ TBAD [16]), to evaluate the contribution of each cohort toward improving model generalisation.

Prior to training, all CTA volumes underwent a consistent preprocessing pipeline to reduce inter-dataset variability. Images were first reoriented into the standard RAS+ coordinate system to ensure consistent anatomical alignment across all datasets. CT intensities were then clipped to a Hounsfield Unit range of -160 to 240 to suppress irrelevant high-density structures such as bone while preserving soft-tissue contrast around the aorta. The clipped intensities were subsequently normalised before training.

During interactive learning, Gaussian-smoothed positive and negative click maps generated by the DeepEdit framework [1] were concatenated with the preprocessed CTA volume, forming a three-channel input for the 3D DynUNet.

2.2 Interactive Segmentation Network

HOLA’s interactive segmentation network is based on the DeepEdit framework [1] and uses a three-dimensional DynUNet backbone. Given a preprocessed CTA volume $I \in \mathbb{R}^{H \times W \times D}$, the segmentation network predicts a binary aorta mask while iteratively incorporating user corrections throughout the refinement process. In contrast to traditional fully automated segmentation, the network output is conditioned on both the underlying CTA volume and the cumulative record of user interactions.

At interaction step t , the network receives a three-channel input consisting of the CTA volume, a positive click map C_t^+ , and a negative click map C_t^- . The positive click map indicates regions that should belong to the aorta, whereas the negative click map highlights incorrectly segmented regions that should be excluded. Adopting the approach from DeepEdit, the system encodes each user interaction as a Gaussian-smoothed volumetric map, accumulating them across subsequent interactions. The segmentation at interaction step t is therefore formulated as

$$S_t = f_\theta(I, C_t^+, C_t^-), \quad t = 0, \dots, T, \quad (1)$$

where f_θ denotes the 3D DynUNet parameterized by θ , and S_t is the predicted binary segmentation after the t -th interaction. At the initial step, no user interaction is available and the click maps are initialized as

$$C_0^+ = C_0^- = \mathbf{0}, \quad (2)$$

allowing the network to produce a fully automatic segmentation. As corrective clicks are added, the click maps are updated and concatenated with the CTA volume, enabling the network to progressively refine the segmentation without retraining.

The segmentation network is trained using a combination of Dice loss and Binary Cross-Entropy (BCE) loss,

$$\mathcal{L}_{seg} = \mathcal{L}_{Dice} + \mathcal{L}_{BCE}, \quad (3)$$

where

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum_i s_i g_i + \epsilon}{\sum_i s_i + \sum_i g_i + \epsilon}, \quad (4)$$

and

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [g_i \log(s_i) + (1 - g_i) \log(1 - s_i)]. \quad (5)$$

Here, s_i and g_i denote the predicted probability and ground-truth label at voxel i , respectively; N is the total number of voxels, and ϵ is a small constant preventing numerical instability. The Dice loss addresses the severe foreground-background imbalance inherent in CTA aorta segmentation, while the BCE term provides stable voxel-wise supervision throughout training.

During training, user interactions are simulated using the DeepEdit strategy, automatically generating positive clicks in false-negative regions and negative clicks in false-positive regions based on slice disagreement. To mimic human behavior, these clicks are sampled stochastically toward the interior of larger error regions rather than deterministically. This process enables a single network to learn both automatic initialization and iterative refinement using identical parameters during inference.

2.3 Reference-Free Quality Prediction

Although the interactive segmentation network progressively refines the predicted aorta mask, it does not indicate whether the current segmentation is sufficiently complete. Therefore, to provide objective guidance during interaction, HOLA incorporates a completeness classifier that estimates the expected segmentation quality without requiring ground-truth annotations during inference.

At each interaction step, the system isolates the foreground voxels by drawing a bounding box around the predicted segmentation, expanding it slightly to maintain surrounding anatomical context. It then crops this region from both the CTA image and the binary mask, resizes them to a standardized $96 \times 96 \times 96$ voxel resolution, then combining into a two-channel input. The completeness classifier consists of four 3D convolutional blocks with channel dimensions of $2 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128$. Each block contains a $3 \times 3 \times 3$ convolution, batch normalization, ReLU activation, and max

pooling. Global average pooling is applied before a fully connected layer with sigmoid activation that predicts a single quality score,

$$q_t = c_\phi(I_\Omega, S_{t,\Omega}) \in [0, 1], \quad (6)$$

where q_t represents the estimated segmentation quality after interaction step t , I_Ω denotes the cropped CTA volume, $S_{t,\Omega}$ is the corresponding predicted mask, and c_ϕ is the completeness classifier.

The classifier is trained using the actual Dice coefficient between the predicted segmentation and the ground truth as the regression target. Training samples are generated by applying the interactive segmentation model to every patient in the training and validation cohorts at interaction steps of 0, 1, 3, 5, 8, and 10 clicks. The held-out test cohort is excluded entirely from classifier training.

The network is optimized using the mean squared error between the predicted and actual Dice scores,

$$\mathcal{L}_{comp} = \frac{1}{N} \sum_{n=1}^N \left(q_t^{(n)} - \text{Dice}(S_t^{(n)}, G^{(n)}) \right)^2, \quad (7)$$

where N is the number of training samples.

During inference, the predicted quality score is compared with a predefined threshold τ . If $q_t < \tau$, additional corrective clicks are recommended; otherwise, the current segmentation is considered sufficiently complete and returned to the user. The threshold is selected using the classifier validation set to balance segmentation quality against the number of user interactions.

2.4 Training Procedure

The interactive segmentation network was trained on the combined CTA datasets described in Section 2.1. HOLA was evaluated using a fixed validation set ($n = 71$) and test set ($n = 72$) across all experiments to ensure consistent and unbiased comparison. Under the progressive multi-dataset training strategy, the training set was expanded from 27 patients at baseline to 49, 82, 116, 158, and finally 214 patients, corresponding to a total of 357 cases across the final training, validation, and test partitions. Every configuration was trained from scratch using a uniform validation protocol, enabling a direct comparison of individual cohorts under consistent evaluation standards.

Training followed the DeepEdit strategy, using randomly cropped $64 \times 128 \times 128$ patches with foreground-biased sampling. During training, 40% of batches were processed without clicks, while the remaining batches received up to three simulated corrective clicks generated from the prediction errors. The three-channel input consisted of the CTA patch and accumulated positive and negative click maps.

The model was optimised using a combined Dice and binary cross-entropy loss, as defined in Equation 3, with the Adam optimiser [7]. An initial learning rate of 1×10^{-3} was used together with cosine annealing. Early stopping with a patience of 15 epochs was applied, and the checkpoint achieving the highest validation Dice score after three corrective clicks was selected for each configuration. Training was performed on the Queen Mary University of London Apocrita High Performance Computing Cluster [6].

2.5 Inference Workflow

During inference, the model generates an initial automatic aorta segmentation from preprocessed CTA data, displaying the result alongside a completeness score. If corrections are needed, users can add positive or negative clicks that are converted into Gaussian interaction maps and fed back into the network to refine the 3D mask. After each refinement step, the updated mask is passed to the completeness classifier, which produces a new quality estimate.

This refinement loop continues with automated guidance, prompting users to stop when the predicted quality reaches the calibrated threshold τ or to continue refining if it remains below this threshold. The final decision to accept the segmentation or perform further correction remains with the user. The final segmentation and quality estimate are then displayed through the Streamlit interface, allowing the user to review the result and perform further corrections when required.

3 Experiments and Results

This section evaluates HOLA through five complementary experiments: overall and cohort-specific segmentation performance, robustness to imperfect user interactions, reference-free quality prediction, multi-dataset training ablations, and comparison with foundation-model baselines. Quantitative evaluation was performed on 72 held-out cases drawn from six CTA cohorts.

3.1 Overall Segmentation Performance

Evaluated on a held-out test set of 72 cases across six CTA cohorts, HOLA achieved an initial automatic DSC of 0.804 (95% CI: 0.7352–0.8692), improving progressively with user guidance to reach 0.851 at one click, 0.861 at five, and 0.868 after ten interactions (95% CI: 0.8099–0.9188). The 0.0633 overall improvement was statistically significant via a Wilcoxon signed-rank test ($p < 0.001$). Final performance reached IoU (Intersection over Union) of 0.820 and HD95 (95th percentile Hausdorff Distance) of 16.27 mm. The sharpest gain occurred at the first interaction (+0.047), after which incremental benefits diminished as major errors were corrected. These results indicate that the greatest benefit of interaction is obtained from the initial corrections, followed by diminishing improvements as segmentation errors are progressively resolved.

3.2 Per-cohort Performance and Generalisation

Table 1 presents the performance of the final HOLA model across the six independent CTA cohorts. Table 1 shows performance across six CTA cohorts. Interactive gains depend on initial quality: SEGA and CIS-UNet improved significantly (+0.142 and +0.075), while Base and Dissection cohorts started high (0.976 and 0.980), leaving little room for improvement. The TBAD cohort offers the key insight: starting lowest at 0.702, interaction improved it to 0.760 (+0.058). This proves interactive correction helps but cannot fully overcome extreme pathological complexity. Leave-one-cohort-out

Table 1. Per-cohort segmentation performance of the final HOLA model on the held-out test set ($N = 72$). $\text{Dice}_{\setminus C}$ is the overall test-set Dice with cohort C excluded, and $\Delta_{\text{LOO}} = \text{Dice}_{\setminus C} - \text{Dice}_{\text{all}}$; a larger Δ_{LOO} indicates a cohort disproportionately dragging down the overall average.

Cohort	Pathology	N	$\text{Dice}_{@0} \uparrow$	$\text{Dice}_{@10} \uparrow$	$\Delta \text{Dice} \uparrow$	$\text{Dice}_{\setminus C} \uparrow$	$\Delta \text{LOO} \uparrow$
Base	Aortic Stenosis	5	0.976	0.977	+0.001	0.860	+0.0082
Dissection [10]	Aortic dissections	8	0.980	0.983	+0.003	0.853	+0.0144
SEGA [11]	Aortic dissections	14	0.775	0.917	+0.142	0.856	+0.0118
CIS-UNet [4]	Type B Aortic Dissections	12	0.845	0.920	+0.075	0.857	+0.0105
AortaSeg-60 [2]	Aortic dissections	7	0.854	0.871	+0.017	0.867	+0.0004
TBAD [16]	Type-B Aortic Dissections	26	0.702	0.760	+0.058	0.929	-0.0612
All cohorts	Combined Test Set	72	0.804	0.868	+0.064	—	—

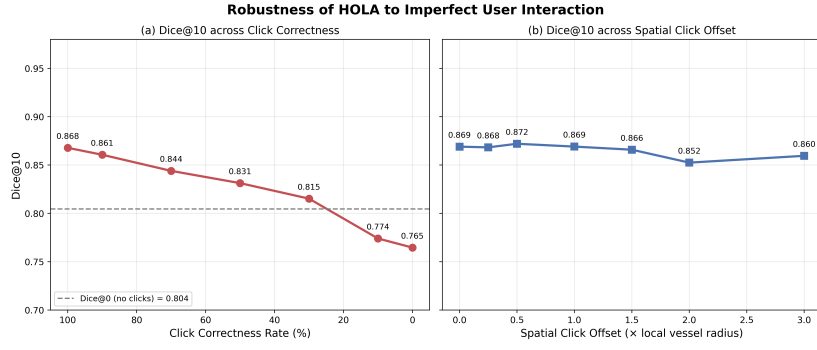


Fig. 3. Robustness of HOLA to spatially imprecise and semantically incorrect user interactions.

analysis (ΔLOO of -0.0612) confirms TBAD serves as a rigorous test domain rather than a negative influence. Ultimately, TBAD highlights the current boundary of interactive refinement and motivates future pathology-aware modelling.

3.3 Robustness to Imperfect User Interaction

The robustness of HOLA to imperfect interaction was evaluated in terms of spatial click precision and semantic correctness. As shown in Figure 3, HOLA remained robust to negative-click displacement of up to $3.0\times$ the local vessel radius, with $\text{Dice}@10$ ranging from 0.852 to 0.872 compared with a 0.868 baseline. In contrast, semantic errors had a stronger impact: at 50% correct decisions, $\text{Dice}@10$ decreased to 0.831, while completely incorrect feedback reduced performance below the initial automatic baseline (0.765 vs. 0.804). These results suggest that HOLA is more tolerant of imprecise click placement than incorrect user decisions.

A pilot evaluation using real user click trajectories across 33 test cases further showed a negligible overall change in Dice (mean $\Delta \text{Dice} = +0.001$). Among 16 multi-click cases, 4 improved, 4 worsened, and 8 showed no change, highlighting the variability of real user interaction compared with the more consistent improvements observed with simulated clicks.

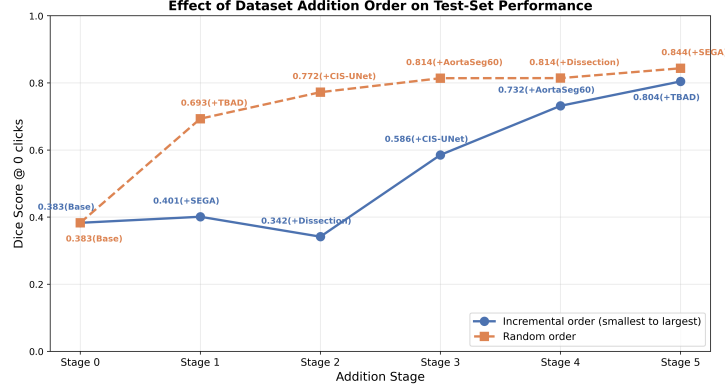


Fig. 4. Multi-cohort training Trajectories under progressive and alternative dataset integration orders.

3.4 Quality Prediction Network Evaluation

Evaluating the prediction network against 432 held-out samples revealed strong alignment with true Dice values $r = 0.953$, and $\text{MAE} = 0.039$. At a calibrated stopping threshold of $\tau = 0.60$, the criterion achieved 95.3% precision, 100% sensitivity, 72.3% specificity, and 95.8% accuracy. Retrospective replay across 72 test patients showed that 61 cases (84.7%) stopped at zero clicks, 4 (5.6%) after one click, and only 7 (9.7%) required the full ten-click budget, decreasing the mean interaction to 1.03 clicks. While incurring a minor mean Dice reduction (0.8466 vs. 0.8690, difference of -0.0224 ; Wilcoxon signed-rank $W = 150$, $p < 0.001$), the 95.3% precision guarantees clinical safety. The quality-guided stopping criterion thus achieves nearly 90% fewer clicks at a modest, statistically confirmed cost to accuracy.

3.5 Ablation Studies

Two complementary ablation studies were conducted to isolate the effects of dataset ordering and training-set size while keeping validation and test distributions fixed.

Impact of Dataset Composition and Integration Order The progressive cohort accumulation strategy was compared with an alternative dataset order: Base $\rightarrow$ TBAD $\rightarrow$ CIS-UNet $\rightarrow$ AortaSeg-60 $\rightarrow$ Dissection $\rightarrow$ SEGA.

As shown in Figure 4, the incremental trajectory exhibited sharp performance fluctuations, such as an initial drop upon introducing the Dissection cohort before recovering, whereas the randomised trajectory maintained a smoother, consistently higher intermediate performance. However, as detailed in Table 2, both strategies ultimately converged to comparable final DSC at ten clicks (0.868 vs. 0.871). This demonstrates that dataset integration order influences intermediate optimisation trajectories but has a limited effect on final interactive segmentation performance once the complete multi-cohort training set is incorporated.

Table 2. Effect of training-set size and dataset integration order on held-out test performance. IoU and HD95 are reported at 10 clicks.

Model	Dice@0↑	Dice@5↑	Dice@10↑	Δ Dice↑	IoU↑	HD95↓
Subset-20	0.629	0.689	0.715	+0.086	0.631	–
Subset-50	0.544	0.678	0.712	+0.168	0.620	–
Subset-100	0.789	0.836	0.849	+0.060	0.795	–
Subset-150	0.754	0.813	0.818	+0.064	0.760	–
Subset-200	0.816	0.870	0.876	+0.060	0.827	14.92
Incremental order	0.804	0.861	0.868	+0.064	0.820	16.27
Random order	0.844	0.868	0.871	+0.027	0.826	12.04

Table 3. Comparison of HOLA with foundation-model baselines on the held-out test set. Dice, IoU, and HD95 are reported at each method’s final result.

Method	Prompt / interaction	Dice↑	IoU↑	HD95 (mm)↓
MedSAM [8]	Bounding box	0.421	0.290	130.48
MedSAM2 [9]	2D clicks, 10 clicks	0.867	0.789	15.88
MedSAM2-3D [12]	2.5D clicks, 10 clicks	0.636	0.497	71.32
HOLA (Ours)	3D clicks, 10 clicks	0.868	0.820	16.27

Impact of Training-set Size As shown in Table 2, final performance increased from 0.715 with 20 patients to 0.876 with 200 patients. Small performance drops at 50 and 150 patients show that having more patients does not automatically guarantee better results. Interestingly, weaker starting models like the 50-patient subset gained the most from user corrections (Δ Dice = +0.168), proving that interactive clicks can easily make up for a weaker initial segmentation.

As summarised in Table 2, the 200-patient multi-source subset achieved the strongest overall performance (Dice@10 of 0.876, IoU of 0.827). Collectively, these findings demonstrate that comprehensive data diversity and sufficient total volume are far more critical for robust vascular segmentation than the precise historical sequence in which datasets are introduced.

3.6 Comparisons with Foundation Baseline Models

Table 3 compares HOLA with foundation models under varying interaction protocols. HOLA uses volumetric corrective clicks, whereas MedSAM2-2D uses slice-wise point prompts. In the matched comparison, HOLA achieved higher Dice than MedSAM2-2D at both zero clicks (0.816 vs. 0.765) and ten clicks (0.880 vs. 0.867). At zero clicks, the median paired Dice difference was +0.057 (95% CI: 0.043–0.090), with a significant Wilcoxon signed-rank test ($W = 613$, $p < 0.001$). After ten clicks, the median paired difference remained +0.042 (95% CI: 0.030–0.048), also significant by Wilcoxon signed-rank testing ($W = 597$, $p < 0.001$). The smaller mean difference at ten clicks (+0.013) indicates that interactive refinement reduced the overall performance gap, while HOLA retained higher performance across the matched cases.

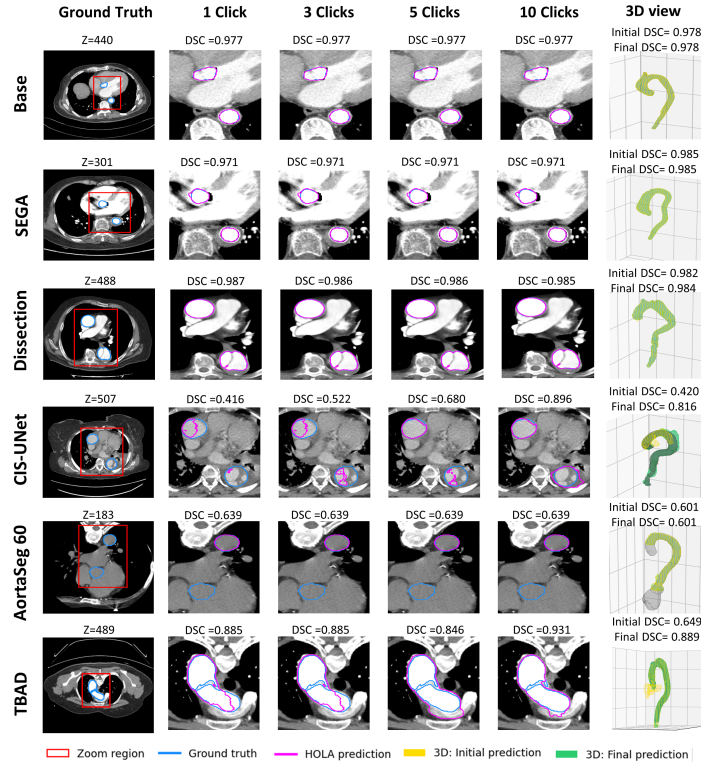


Fig. 5. Representative examples of interactive segmentation refinement across the six CTA cohorts.

3.7 Cross-Cohort Qualitative Evaluation of Interactive Refinement

Figure 5 illustrates representative cohort-specific refinement behaviours across the interaction sequence. In several cases, including the Baseline, SEGA, and Dissection examples, the segmentation remained stable after the initial prediction because the prediction network had already identified the result as sufficient quality at the corresponding iteration. The absence of further changes after this point therefore reflects successful quality-guided stopping rather than a failure to improve. In contrast, the CIS-UNet example required further refinement, with 2D Dice increasing from 0.416 to 0.896 and volumetric Dice from 0.420 to 0.816, demonstrating the benefit of corrective interaction when substantial initial errors were present. AortaSeg-60 showed limited improvement, while TBAD demonstrated substantial volumetric improvement despite its challenging pathology. Overall, the examples demonstrate that HOLA can distinguish between cases requiring further correction and cases where the current segmentation is already sufficiently accurate, reducing unnecessary user interaction.

4 Conclusion

This paper presents HOLA, a 3D human-AI collaborative framework for interactive aortic segmentation from CTA, combining volumetric corrective interaction with reference-free quality-guided feedback. Results demonstrated that interaction effectively refines challenging segmentations, while quality-aware stopping substantially cuts down the required workload while maintaining high accuracy. Furthermore, evaluation across diverse datasets highlights that while multi-dataset training improves generalisation, severe pathology like TBAD remains challenging, underscoring the value of integrating proactive refinement with quality-aware estimation to streamline clinical workflows.

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